

# **ECO310 PS3 Q1 Teaching Guide: Cost Functions & Monopoly**

WithAI Research

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# ECO310 Problem Set 3, Question 1 Teaching Guide

## Cost Functions & Monopoly Pricing - Comprehensive Lecture Review

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## 1. Overview: What You Need to Know

This guide covers **two major topics** from Lectures 11 & 12 that you missed:

1. **Question 1A:** Deriving cost functions from production technology
2. **Question 1B:** Monopoly profit maximization

### Big Picture

We're transitioning from **consumer theory** (how consumers make choices) to **producer theory** (how firms make production decisions). This involves:

- Understanding how firms produce goods (production functions)
  - How much it costs to produce goods (cost functions)
  - How firms price their products (perfect competition vs. monopoly)
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## 2. Part 1: Production Functions and Cost Functions (Question 1A)

### 2.1 What is a Production Function?

**Definition:** A **production function** shows the relationship between inputs (like labor, capital) and outputs (goods produced).

**General form:**  $q = f(L, K)$

Where:

- $q$  = quantity of output produced
- $L$  = labor input (number of workers or hours)
- $K$  = capital input (machines, equipment)

## 2.2 Common Production Functions You'll See

1. **Linear:**  $q = aL + bK$  (constant returns, each input adds constant amount)
2. **Square root:**  $q = \sqrt{L}$  or  $q = 1 + \sqrt{L}$  (**this is Question 1A!**)
3. **Cobb-Douglas:**  $q = L^\alpha K^\beta$  (most common in economics)
4. **Quadratic:**  $q = L^2$  or  $q = K^2$

## 2.3 From Production to Cost: The Key Steps

### The Process for Deriving Cost Functions

**Step 1:** Start with production function  $q = f(L, K)$

**Step 2: Invert** to get input in terms of output

- Solve for  $L$  or  $K$  as a function of  $q$
- Example: If  $q = \sqrt{L}$ , then  $L = q^2$

**Step 3:** Multiply by input prices

- If labor costs  $w$  per unit: Cost =  $w \cdot L$
- If capital costs  $r$  per unit: Cost =  $r \cdot K$
- Total cost:  $C(q) = wL(q) + rK(q)$

**Step 4:** Identify fixed vs. variable costs

- Fixed costs ( $F$ ): Don't change with  $q$  (rent, overhead)
- Variable costs ( $VC(q)$ ): Change with  $q$  (labor, materials)
- $C(q) = F + VC(q)$

## 2.4 Worked Example: Similar to Question 1A

**Setup:** A firm has production function  $q = \sqrt{L}$  where  $L$  is labor. Each unit of labor costs  $w = \$4$ .

**Question:** Find the cost function  $C(q)$ .

**Solution:**

**Step 1:** Production function is  $q = \sqrt{L}$

**Step 2:** Invert to get  $L$  in terms of  $q$ :

- $q = \sqrt{L}$
- $q^2 = L$  (square both sides)

So we need  $L = q^2$  units of labor to produce  $q$  units of output.

**Step 3:** Multiply by wage rate:

- $C(q) = w \cdot L$
- $C(q) = 4 \cdot q^2$
- $C(q) = 4q^2$

**Answer:**  $C(q) = 4q^2$

**Interpretation:** To produce 1 unit costs \$4. To produce 2 units costs \$16. To produce 3 units costs \$36. Costs increase *quadratically* with output!

## 2.5 Important Cost Function Terminology

Once you have  $C(q)$ , you can derive several important concepts:

### Cost Function Components

1. **Total Cost:**  $C(q) = F + VC(q)$ 
  - $F$  = fixed costs (constant, independent of  $q$ )
  - $VC(q)$  = variable costs (depends on  $q$ )
2. **Marginal Cost:**  $MC(q) = C'(q) = dC/dq$ 
  - Cost of producing one more unit
  - **Key for profit maximization!**
3. **Average Cost:**  $AC(q) = C(q)/q$ 
  - Cost per unit
4. **Average Variable Cost:**  $AVC(q) = VC(q)/q$ 
  - Variable cost per unit
  - Important for shutdown decisions (firms shut down if  $p < \min AVC$ )

## 2.6 Question 1A: Step-by-Step Approach

### Problem 1A

A firm produces output according to  $q = 1 + \sqrt{L}$  where  $q$  is output and  $L$  is labor. Each labor unit costs \$2. Find the cost function  $C(q)$ .

### Your Turn - Follow These Steps:

**Step 1:** Write down the production function

$$q = 1 + \sqrt{L}$$

**Step 2:** Solve for  $L$  in terms of  $q$  (invert the function)

$$q = 1 + \sqrt{L} \quad q - 1 = \sqrt{L} \quad (\text{subtract 1 from both sides}) \quad (q-1)^2 = L \quad (\text{square both sides})$$

$$\text{So } L(q) = (q-1)^2$$

**Step 3:** Multiply by wage rate ( $w = 2$ )

$$C(q) = w \cdot L(q) \quad C(q) = 2 \cdot (q-1)^2 \quad C(q) = 2(q^2 - 2q + 1) \quad C(q) = 2q^2 - 4q + 2$$

### Answer to 1A

$$C(q) = 2q^2 - 4q + 2 \text{ or equivalently } C(q) = 2(q-1)^2$$

### Components:

- Fixed cost:  $F = 2$  (the constant term when  $q = 0$ , though technically  $C(0) = 2$ )
- Variable cost:  $VC(q) = 2q^2 - 4q$  (the part that depends on  $q$ )
- Marginal cost:  $MC(q) = C'(q) = 4q - 4 = 4(q-1)$

### ■ Common Mistake

Don't forget to square **both terms** when you have  $(q-1)^2$ :

✓ **Correct:**  $(q-1)^2 = q^2 - 2q + 1$

✗ **Wrong:**  $(q-1)^2 = q^2 - 1$  or  $q^2 + 1$

## 3. Part 2: Monopoly and Profit Maximization (Question 1B)

### 3.1 Perfect Competition vs. Monopoly: The Big Difference

| Feature          | Perfect Competition                                | Monopoly                                     |
|------------------|----------------------------------------------------|----------------------------------------------|
| Number of firms  | Many small firms                                   | One firm (the monopolist)                    |
| Price-taking?    | Yes - firms take price as given                    | No - firm sets price                         |
| Demand curve     | Horizontal (perfectly elastic) at market price $p$ | Downward-sloping market demand $P^D(Q)$      |
| Marginal Revenue | $MR = p$ (constant)                                | $MR = P^D(Q) + Q \cdot dP^D/dQ$ (decreasing) |
| Profit max       | $p = MC$                                           | $MR = MC$                                    |

### 3.2 Why Does Market Structure Matter?

#### Perfect Competition Example

Imagine you're selling pencils at a farmer's market. There are 100 other vendors selling identical pencils.

- If everyone sells at \$1, you must also sell at \$1
- If you charge \$1.10, nobody buys from you
- You're a **price-taker** - you take the market price as given
- Your revenue from selling  $q$  units:  $R(q) = p \cdot q = 1 \cdot q$
- Marginal revenue:  $MR = dR/dq = 1$  (constant!)

#### Monopoly Example

Now imagine you're Apple in 2010, the only company selling tablets.

- You control the entire market
- If you want to sell more units, you must **lower the price**
- The market demand curve tells you: at price  $P$ , consumers want  $Q^D(P)$  units
- You must decide: high price + low quantity, or low price + high quantity?
- Your revenue:  $R(Q) = P^D(Q) \cdot Q$  (price depends on quantity!)

## 3.3 Understanding Demand Curves

### Two Ways to Express the Same Relationship

**Demand curve:**  $Q^D(P)$  tells you quantity demanded at price  $P$

- Example:  $Q^D(P) = 200 - 5P$
- "At price  $P = 10$ , consumers demand  $Q = 200 - 50 = 150$  units"

**Inverse demand curve:**  $P^D(Q)$  tells you the price at which you can sell  $Q$  units

- Solve for  $P$ :  $Q = 200 - 5P \Rightarrow 5P = 200 - Q \Rightarrow P = 40 - Q/5$
- Example:  $P^D(Q) = 40 - Q/5$
- "To sell  $Q = 150$  units, you must charge price  $P = 40 - 30 = 10$ "

**For monopoly problems, we typically use inverse demand  $P^D(Q)$  because the monopolist chooses quantity  $Q$  and the price is determined by the demand curve.**

## 3.4 Monopolist's Revenue and Marginal Revenue

### Monopolist's Revenue

**Total Revenue:**

$$R(Q) = P^D(Q) \cdot Q$$

The monopolist sells  $Q$  units at price  $P^D(Q)$ .

### Marginal Revenue

Revenue from selling one more unit:

$$MR(Q) = dR/dQ = d/dQ[P^D(Q) \cdot Q]$$

Using the product rule:

$$MR(Q) = P^D(Q) + Q \cdot dP^D/dQ$$

Since demand is downward-sloping,  $dP^D/dQ < 0$ , so:

**$MR(Q) < P^D(Q)$  for a monopolist**

### ■ Critical Insight: Why is $MR < P$ for a monopolist?

When a monopolist sells one more unit:

1. **Gain:** Revenue from selling that unit at price  $P$  ( $+P$ )
2. **Loss:** To sell more, price must drop, reducing revenue on *all previous units* ( $-Q \cdot \Delta P$ )

**Example:** If you sell 100 tablets at \$500 each ( $R = \$50,000$ ), and to sell 101 tablets you must drop price to \$499:

- Gain from 101st tablet:  $+\$499$
- Loss on first 100 tablets:  $-100 \times \$1 = -\$100$
- Net marginal revenue:  $\$499 - \$100 = \$399 < \$499$  (the price!)

## 3.5 Monopoly Profit Maximization

### The Monopolist's Problem

The monopolist wants to maximize profit:

$$\Pi(Q) = R(Q) - C(Q) = P^D(Q) \cdot Q - C(Q)$$

**First Order Condition:**

$$d\Pi/dQ = 0 \quad \blacksquare \quad MR(Q) = MC(Q)$$

### Steps to Solve:

1. Write out profit function:  $\Pi(Q) = P^D(Q) \cdot Q - C(Q)$
2. Take derivative with respect to  $Q$ :  $d\Pi/dQ = MR - MC$
3. Set equal to zero:  $MR = MC$
4. Solve for optimal quantity  $Q^*$
5. Plug  $Q$  into inverse demand to find price:  $P = P^D(Q^*)$

## 3.6 Calculating Marginal Revenue: The Formula

### MR Formula for Linear Demand

If inverse demand is **linear**:  $P^D(Q) = a - bQ$

Then marginal revenue is:  **$MR(Q) = a - 2bQ$**

**Rule of thumb:** MR has the **same intercept** but **twice the slope** of inverse demand!

**Why?**

$$R(Q) = P^D(Q) \cdot Q = (a - bQ) \cdot Q = aQ - bQ^2 \quad MR(Q) = dR/dQ = a - 2bQ$$

### Example: Computing MR

**Inverse demand:**  $P^D(Q) = 200 - 5Q$

**Marginal revenue:**

$$R(Q) = P^D(Q) \cdot Q = (200 - 5Q) \cdot Q = 200Q - 5Q^2 \quad MR(Q) = dR/dQ = 200 - 10Q$$

Or use the shortcut:  $P = 200 - 5Q \quad \blacksquare \quad MR = 200 - 10Q$  (same intercept, double the slope!)



## 3.7 Worked Example: Monopoly Profit Maximization

### Full Monopoly Problem

**Setup:** A monopolist has cost function  $C(Q) = Q^2 + 1$  and faces inverse demand  $P^D(Q) = 400 - Q$ .

**Question:** Find the profit-maximizing quantity and price.

### Solution:

**Step 1:** Write profit function

$$\Pi(Q) = R(Q) - C(Q) = P^D(Q) \cdot Q - C(Q) = (400 - Q) \cdot Q - (Q^2 + 1) = 400Q - Q^2 - Q^2 - 1 = 400Q - 2Q^2 - 1$$

**Step 2:** Find marginal revenue and marginal cost

$$MR(Q) = dR/dQ = d/dQ[(400 - Q) \cdot Q] = 400 - 2Q \quad MC(Q) = dC/dQ = 2Q$$

**Step 3:** Set  $MR = MC$

$$400 - 2Q = 2Q \quad 400 = 4Q \quad Q^* = 100$$

**Step 4:** Find price by plugging into demand

$$P^* = P^D(Q^*) = 400 - 100 = 300$$

**Answer:** The monopolist sells  $Q = 100$  units at price  $P = \$300$ .

**Profit check:**

$$\Pi(100) = 300 \cdot 100 - (100^2 + 1) = 30,000 - 10,001 = 19,999$$

The monopolist makes \$19,999 in profit!

## 3.8 Question 1B: Step-by-Step Approach

### Problem 1B

Suppose the firm is a monopolist and faces demand curve  $Q^D(P) = 200 - 5P$ . The cost function is  $C(q) = 2(q-1)^2$  from part (a). How much will the monopolist sell, and at what price?

### Your Turn - Follow These Steps:

**Step 1:** Convert demand to inverse demand

$$Q^D(P) = 200 - 5P \quad 5P = 200 - Q \quad P^D(Q) = 40 - Q/5$$

**Step 2:** Write profit function

$$\begin{aligned} \Pi(Q) &= P^D(Q) \cdot Q - C(Q) = (40 - Q/5) \cdot Q - 2(Q-1)^2 = 40Q - Q^2/5 - 2(Q^2 - 2Q + 1) = 40Q - Q^2/5 - 2Q^2 + 4Q - 2 \\ &= 44Q - Q^2/5 - 2Q^2 - 2 \end{aligned}$$

Simplify the  $Q^2$  terms:

$$-Q^2/5 - 2Q^2 = -Q^2/5 - 10Q^2/5 = -11Q^2/5$$

So:

$$\Pi(Q) = 44Q - 11Q^2/5 - 2$$

**Step 3:** Find MR and MC

$$MR(Q) = d/dQ[(40 - Q/5) \cdot Q] = 40 - 2Q/5 \quad MC(Q) = dC/dQ = d/dQ[2(Q-1)^2] = 2 \cdot 2(Q-1) = 4(Q-1) = 4Q - 4$$

**Step 4:** Set MR = MC

$$40 - 2Q/5 = 4Q - 4 \quad 40 + 4 = 4Q + 2Q/5 \quad 44 = 20Q/5 + 2Q/5 \quad 44 = 22Q/5 \quad 220 = 22Q \quad Q^* = 10$$

**Step 5:** Find price

$$P^* = P^D(Q^*) = 40 - 10/5 = 40 - 2 = 38$$

### Answer to 1B

$Q = 10$  units,  $P = \$38$

The monopolist will produce and sell 10 units at a price of \$38 per unit.

**Check:**

- Revenue =  $38 \times 10 = \$380$
- Cost =  $C(10) = 2(10-1)^2 = 2(81) = \$162$
- Profit =  $380 - 162 = \$218$

## 4. Additional Concepts: Full Context

### 4.1 Why Do We Care About Market Structure?

#### Economic Efficiency

**Perfect competition** leads to efficient outcomes:

- Price = Marginal cost ( $P = MC$ )
- Consumer surplus + Producer surplus is maximized
- No deadweight loss

**Monopoly** creates inefficiency:

- Price > Marginal cost ( $P > MC$ )
- Monopolist restricts quantity to raise price
- Creates **deadweight loss** (lost welfare)
- Transfers surplus from consumers to the monopolist

### 4.2 Types of Costs: Complete Summary

| Cost Type             | Formula            | Meaning                                              |
|-----------------------|--------------------|------------------------------------------------------|
| Total Cost            | $C(q) = F + VC(q)$ | Total cost of producing $q$ units                    |
| Fixed Cost            | $F$                | Costs that don't change with output (rent, machines) |
| Variable Cost         | $VC(q)$            | Costs that change with output (labor, materials)     |
| Marginal Cost         | $MC(q) = C'(q)$    | Cost of producing one more unit                      |
| Average Total Cost    | $AC(q) = C(q)/q$   | Cost per unit                                        |
| Average Variable Cost | $AVC(q) = VC(q)/q$ | Variable cost per unit                               |
| Average Fixed Cost    | $AFC(q) = F/q$     | Fixed cost per unit (decreases as $q$ increases)     |

### 4.3 Profit Maximization: All Market Structures

#### Universal Profit Maximization Rule

**All firms** (competitive or monopolist) maximize profit by producing where:

$$MR(Q) = MC(Q)$$

The difference is **what MR equals**:

- **Perfect competition:**  $MR = P$  (constant)
- **Monopoly:**  $MR = P^D(Q) + Q \cdot dP^D/dQ < P$  (decreasing)

## 5. Common Mistakes and How to Avoid Them

### Mistake 1: Forgetting to Invert Production Function

#### ■ Don't Do This

**Wrong:**  $q = 1 + \sqrt{L}$  ■  $C(q) = 2(1 + \sqrt{q})$

You can't just replace  $L$  with  $q$ ! You must **solve for  $L$  first**.

#### ✓ Correct Approach

$$q = 1 + \sqrt{L} \quad \blacksquare \quad L = (q-1)^2 \quad \blacksquare \quad C(q) = 2(q-1)^2$$

### Mistake 2: Using Price Instead of MR for Monopolist

#### ■ Don't Do This

**Wrong:** "Monopolist maximizes profit where  $P = MC$ "

This is the **perfect competition** rule! For monopolists,  $MR \neq P$ .

#### ✓ Correct Approach

Monopolist maximizes where  $MR = MC$ , where:

$$MR = P^D(Q) + Q \cdot \frac{dP^D}{dQ} < P^D(Q)$$

### Mistake 3: Confusing Demand and Inverse Demand

#### ■ Don't Do This

Given  $Q^D(P) = 200 - 5P$ :

**Wrong:** "Revenue =  $(200 - 5P) \cdot P$ "

This mixes up the variable! If you're using  $Q$  as your choice variable (which monopolists do), you need inverse demand.

#### ✓ Correct Approach

1. Convert to inverse demand:  $P^D(Q) = 40 - Q/5$
2. Revenue:  $R(Q) = P^D(Q) \cdot Q = (40 - Q/5) \cdot Q$

### Mistake 4: Not Simplifying Before Taking Derivatives

#### ■ Don't Do This

**Wrong:** Taking derivative of  $\Pi(Q) = (40 - Q/5) \cdot Q - 2(Q-1)^2$  directly without expanding.

This leads to messy product rule and chain rule applications.

#### ✓ Correct Approach

Expand everything first:

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$$\Pi(Q) = 40Q - Q^2/5 - 2Q^2 + 4Q - 2 = 44Q - 11Q^2/5 - 2$$

Now taking the derivative is easy!

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## 6. Practice Problems

### Practice Problem 1: Cost Functions

**Problem:** A firm has production function  $q = 2\sqrt{L}$  where labor costs \$9 per unit. Find  $C(q)$ .

#### Solution Space

Step 1: Invert production function Step 2: Multiply by wage Answer:

### Practice Problem 2: Monopoly

**Problem:** A monopolist has  $C(Q) = 3Q^2$  and faces demand  $Q^D(P) = 100 - 2P$ . Find optimal  $Q$  and  $P$ .

#### Solution Space

Step 1: Convert to inverse demand Step 2: Find MR and MC Step 3: Set  $MR = MC$  and solve Step 4: Find price Answer:

### Practice Problem 3: Comparing Outcomes

**Problem:** Using the same demand and cost as Practice Problem 2, what would  $Q$  and  $P$  be under perfect competition?

#### Solution Space

Perfect competition rule:  $P = MC$  Answer:

# Solutions to Practice Problems

## Solution 1

**Given:**  $q = 2\sqrt{L}$ ,  $w = 9$

**Step 1:** Invert production function

$$q = 2\sqrt{L} \quad q/2 = \sqrt{L} \quad (q/2)^2 = L \quad L = q^2/4$$

**Step 2:** Multiply by wage

$$C(q) = 9 \cdot q^2/4 = 9q^2/4$$

**Answer:**  $C(q) = 9q^2/4$

## Solution 2

**Given:**  $C(Q) = 3Q^2$ ,  $Q^D(P) = 100 - 2P$

**Step 1:** Inverse demand

$$Q = 100 - 2P \quad \blacksquare \quad P^D(Q) = 50 - Q/2$$

**Step 2:** Find MR and MC

$$MR = 50 - Q \quad (\text{double the slope}) \quad MC = 6Q$$

**Step 3:** Set MR = MC

$$50 - Q = 6Q \quad 50 = 7Q \quad Q^* = 50/7 \approx 7.14$$

**Step 4:** Find price

$$P^* = 50 - (50/7)/2 = 50 - 25/7 = 325/7 \approx 46.43$$

**Answer:**  $Q \approx 7.14$  units,  $P \approx \$46.43$

## Solution 3

**Perfect competition:** Set  $P = MC$

$$P = 50 - Q/2 \quad (\text{from inverse demand}) \quad MC = 6Q \quad 50 - Q/2 = 6Q \quad 50 = 6Q + Q/2 = 13Q/2 \quad Q_{pc} = 100/13 \approx 7.69 \\ P_{pc} = 6 \cdot (100/13) = 600/13 \approx 46.15$$

**Comparison:**

- Monopoly:  $Q = 7.14$ ,  $P = 46.43$
- Perfect competition:  $Q = 7.69$ ,  $P = 46.15$
- Monopolist produces less and charges more!

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## 7. Quick Reference: Formulas You Need

### Essential Formulas Cheat Sheet

#### Cost Functions

- $C(q) = F + VC(q)$  (total = fixed + variable)
- $MC(q) = C'(q)$  (marginal cost)
- $AC(q) = C(q)/q$  (average cost)

#### Revenue

- $R(Q) = P \cdot Q$  (revenue = price  $\times$  quantity)
- For monopolist:  $R(Q) = P^D(Q) \cdot Q$  (price depends on Q!)

#### Marginal Revenue

- Perfect competition:  $MR = P$  (constant)
- Monopoly:  $MR = P^D(Q) + Q \cdot dP^D/dQ$
- For linear demand  $P = a - bQ$ :  $MR = a - 2bQ$

#### Profit

- $\Pi(Q) = R(Q) - C(Q)$
- Maximize:  $d\Pi/dQ = 0$  ■  $MR = MC$

#### Inverse Demand

- Given  $Q^D(P) = a - bP$ , inverse is  $P^D(Q) = a/b - Q/b$
- Example:  $Q = 200 - 5P$  ■  $P = 40 - Q/5$

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## Final Checklist for Problem Set 3, Question 1

### ✓ Question 1A:

- Invert production function  $q = 1 + \sqrt{L}$  to get  $L = (q-1)^2$
- Multiply by wage:  $C(q) = 2(q-1)^2 = 2q^2 - 4q + 2$
- Check by verifying  $MC = C'(q) = 4q - 4$

### ✓ Question 1B:

- Convert demand to inverse:  $P^D(Q) = 40 - Q/5$
  - Find MR:  $MR = 40 - 2Q/5$
  - Find MC from part (a):  $MC = 4Q - 4$
  - Set  $MR = MC$  and solve:  $Q^* = 10$
  - Find price:  $P^* = 40 - 2 = 38$
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# You're ready to solve the problem!

Good luck, and remember: take it step by step!